Naga Sindhuri Munjila

Data Engineer

CONTACT

(408)-673-1227

sindhumunjila@gmail.com

[LinkedIN]()

EDUCATION

Arizona State University, Arizona, USA

Masters in Information Technology, Jan 2023 to Dec 2024

KEY SKILLS

Big Data Technologies: HDFS, YARN, MapReduce, Hive, Pig, Impala, Sqoop, Storm, Flume, Spark, Apache Kafka, Zookeeper, Ambari, Oozie, MongoDB, Cassandra, Mahout, Puppet, Avro, Parquet, Snappy, Falcon.

NO SQL Databases: Postgres, HBase, Cassandra, MongoDB, Amazon DynamoDB, Redis

Hadoop Distributions: Cloudera (CDH3, CDH4, and CDH5), Hortonworks, MapR, and Apache.

Languages: Scala, Python, R, XML, XHTML, HTML, AJAX, CSS, SQL, PL/SQL, HiveQL, Shell Scripting

Data Modeling & Architecture: Star Schema, Snowflake Schema, Erwin, ER/Studio

Cloud Computing Tools: Amazon AWS, (S3, EMR, EC2, Lambda, VPC, Route 53, Cloud Watch, CloudFront), Microsoft Azure, GCP

Databases: Teradata Snowflake, Microsoft SQL Server, MySQL, DB2, PL/SQL, PostgreSQL & Oracle

Build Tools: Jenkins, Maven, Ant, Log4j

Business Intelligence Tools: Tableau, Power BI

ETL Tools: Azure Data Factory, Talend, Pentaho, Informatica, SSIS


OBJECTIVE

Results-driven Data Engineer with 6+ years of extensive experience in designing, building, and maintaining scalable data infrastructure on cloud platforms, including Azure, AWS, and Google Cloud (GCP). Adept at leveraging cloud-native services to optimize ETL processes, data pipelines, and storage solutions, ensuring high performance and reliability.

PROFILE SUMMARY

- 6+ years of expertise designing, developing, and executing data pipelines and data **lake** requirements in numerous companies using the Big Data Technology stack, **Python**, **PL/SQL**, **SQL**, **REST APIs**, **GraphQL** and the **Azure** cloud platform.
- Experience in Evaluation/design/development/deployment of additional technologies and automation for managed services on **S3**, **Lambda**, **Athena**, **EMR**, **Kinesis, SQS, SNS, CloudWatch, Data Pipeline**, **Redshift**, **Dynamo DB, AWS Glue**, Aurora DB, **RDS**, **EC2**. Experience in migrating on premise to **Windows Azure** using **Azure** Site Recovery and **Azure** backups.
- **ETL pipelines** in and out of data warehouse using **Python** with **Pandas** and **Spark.** Strong Experience using **AWS** and **Azure** Data services and creating the most Efficient Data Pipelines using optimized Solutions.
- Implemented Big Data solutions using Hadoop technology stack, including **PySpark**, **Hive**, **Sqoop**, **Avro** and **Thrift**. Working knowledge of **HDFS, Kafka, Map Reduce, Spark, PIG, HIVE, Sqoop, HBase, Flume, and Apache ZooKeeper** as tools for designing and deploying end-to-end big data ecosystems.
- Develop batch processing solutions by using **Azure Data Factory** and **Azure** Data bricks. Strong Experience using **AWS** and **Azure** Data services and creating the most Efficient Data Pipelines using optimized Solutions.
- Created **Python** code to collect data from **HBase** and develop the **PySpark** implementation of the solution.
- Experience in automating day-to-day activities by using **Windows PowerShell.**
- Experience working in environments using **Agile** (**SCRUM**) and Test-Driven development (**TDD**) methodologies.
- Architect & implement medium to large scale BI solutions on **Azure** using **Azure** Data Platform services (**Azure Data Lake**, **Azure Data Factory**, **Data Lake Analytics**, and Stream Analytics).
- Practical experience with **Python** and **Apache Airflow** to create, schedule, and monitor workflows. Proficient in utilizing **Kubernetes** and **Docker** for designing and implementing data pipelines.
- Experience working with Front end technologies like **Html**, **CSS**, **JS**, **ReactJS**.
- Experience with **Snowflake** cloud data warehouse and **AWS S3** bucket for integrating data from multiple source system which include loading nested **JSON** formatted data into snowflake table. Expertise in building **CI/CD** on **AWS** environment using **AWS** Code Commit, Code Build, Code Deploy and Code Pipeline.

EMPLOYMENT HISTORY

Farmers Insurance Group || Los Angeles, California, USA
Azure Data Engineer || Jun 2024 - Current

Description: Farmers Insurance Group is an American insurer group of vehicles, homes and small businesses and also provides other insurance and financial services products. I develop and manage ETL/ELT processes to ingest data from various sources. Implement and manage Azure data storage solutions such as Azure SQL Database, Azure Data Lake Storage, and Cosmos DB.

Responsibilities:

- Looked into existing **Scala spark** processing and maintained, enhanced the jobs.
- Strong knowledge of **ETL** best practices and experience designing and implementing **ETL** workflows using Talend.
- Load and transform large sets of structured, semi structured, and unstructured data using **Hadoop/Big Data concepts.**
- Enabling monitoring and **Azure** log analytics to alert support team on usage and stats of the daily runs.
- Develop **Spark** streaming application tread raw packet data from **Kafka** topics, format it to **JSON** and push back to **Kafka** for future use case's purpose. Performed **ETL** to move the data from source system to destination systems and worked on the Data warehouse. Involved in database migration methodologies and integration conversion solutions to convert legacy **ETL** processes into **Azure** Synapse compatible architecture.
- Processed the data efficiently with in **Azure** Databricks and visualizing insights through **Tableau** dashboards
- Created Data tables utilizing **PyQt** to display customer and policy information and add, delete, update customer records.
- Used **Python** for data validation and analysis purposes.
- Utilized **Elasticsearch** and Kibana for indexing and visualizing the real-time analytics results, enabling stakeholders to gain actionable insights quickly. Involved in various phases of Software Development Lifecycle (**SDLC**) of the application, like gathering requirements, design, development, deployment, and analysis of the application.
- Designed and implemented **Star and Snowflake Schemas** in Snowflake, enabling efficient reporting and BI analytics and Utilized **Erwin** for data modeling, ensuring clear metadata management and documentation of data relationships.
- Used **Azure Data Factory** to ingest data from log files and business custom applications, processed data on Data bricks per day-to-day requirements, and loaded them to **Azure** Data Lakes.
- Worked on migration of data from On - prem **SQL** Server to Cloud databases (Azure Synapse **Analytics** (DW) & **Azure** SOL DB). Build and deployed the code Artefacts into the respective environments in the Confidential **Azure** cloud.
- Used **Jira** for ticketing and tracking issues and **Jenkins** for continuous integration and continuous deployment.
- Controlling and granting database access and migrating on premise databases to **Azure** data **lake** store using **Azure** Data **Factory**. Deployed models as **Python** package, as **API** for backend integration and as services in a **Microservices** architecture with a **Kubernetes** orchestration layer for the **Dockers containers.**
- Skilled in monitoring servers using **Nagios, Cloud watch** and using **ELK Stack- Elastic search** and **Kibana.**
- Achieved performance improvement on Synapse loading by implementing a dynamic partition switch.
- Develop metrics based on **SAS** scripts on legacy system, migrating metrics to **snowflake** (**Google Cloud**). Implemented a continuous delivery (**CI/CD) pipeline** with **Docker** for custom application images in the cloud using **Jenkins**.

Environment: API, Azure, Azure Data Lake, CI/CD, Data Factory, Docker, Elasticsearch, ETL, Erwin, Factory, GraphQL, Java, Jenkins, Jira, JS, Kafka, Kubernetes, PySpark, Python, SAS, Scala, Spark, SQL, Tableau

Edwards Lifesciences || Irvine, California, USA
AWS Data Engineer || Apr 2023 - May 2024

Description: Edwards Lifesciences is an American medical technology company. Ensured data solutions comply with regulatory requirements and security best practices. Implemented the data encryption, access controls, and auditing mechanisms using AWS IAM, KMS, and other security tools.

Responsibilities:

- Designed **SSIS** control flow tasks for orchestrating the sequence and logic of **ETL** processes, such as conditional branching, looping, and error handling. Exploring with the **PySpark** to improve the performance and optimization of the existing algorithms in Hadoop using **Spark** Context, Spark-SQL, Data Frame, and Pair RDD's.
- Integrated **AWS DynamoDB** using **AWS Lambda** to store the values of items and backup the **DynamoDB** streams.

- Have worked on partition of **Kafka** messages and setting up the replication factors in **Kafka Cluster**.
- Developed **Python** code to gather the data from **HBase** (Cornerstone) and designs the solution to implement using **PySpark**. Automated and monitored **AWS** infrastructure with Terraform for high availability and reliability, reducing infrastructure management time by 90% and improving system uptime.
- Designing the business requirement collection approach based on the project scope and SDLC methodology.
- Created Complex Stored Procedures, Slow Changing Dimension Type 2, Triggers, Functions, Tables, Views and other T **SQL** code and **SQL** joins to ensure efficient data retrieval.
- Built and configured **Jenkins** slaves for parallel job execution. Installed and configured **Jenkins** for continuous integration and performed continuous deployments. Working on migrating Data to the cloud (Snowflake and AWS) from the legacy data warehouses and developing the infrastructure.
- Integrated **Kubernetes** with cloud-native services, such as **AWS** EKS and **GCP** GKE, to leverage additional scalability and managed services. Developed Kibana Dashboards based on the Log stash data and Integrated different source and target systems into **Elasticsearch** for near real time log analysis of monitoring End to End transactions.s
- Proficient in using **Snowflake** utilities, Snow **SQL**, Snow Pipe, and applying Big Data modeling techniques using **Python**.
- Built **Snowflake-based** data warehouses for patient monitoring systems, ensuring HIPAA compliance. Implemented dimensional modeling (**Star Schema**) in Snowflake for improved analytical performance on healthcare datasets.
- Used **ER/Studio** to create and maintain catalogs for tracking data lineage and schema changes.
- Used **Spark** for interactive queries, processing of streaming data and integration with popular **NoSQL database** for huge volume of data. Implementedusageof**AmazonEMR**forprocessing**BigData**acrossa**HadoopCluster**ofvirtualservers on **Amazon Elastic Compute Cloud** (**EC2**) and **Amazon Simple Storage Service(S3)**
- Involved in daily **Scrum meetings** to discuss the development /progress and was active in making scrum meetings more productive. Seamlessly worked on **Python** to build **data pipelines** after the data got loaded from **Kafka.**
- Used **Kafka** Streams to Configure **Spark Streaming** to get information and then store it in **HDFS**.
- Worked on loading data into **Spark RDD's**, perform advanced procedures like text analytics using in- memory data computation capabilities of **Spark** to generate the Output response. Created **AWS Lambda** functions and assigned **IAM** roles to schedule python scripts using **CloudWatch Triggers** to support the infrastructure needs (**SQS**, Event Bridge, **SNS**)
- Involved in converting **MapReduce** programs into **Spark** transformations using **Spark RDD's** using **Scala** and **Python**.
- Integrated **Kafka- Spark streaming** for high efficiency through put and reliability.
- Conducted **ETL** Data Integration, Cleansing, and Transformations using **AWS glue Spark script.**
- Created datasets from **S3** using **AWS Athena** and created Visual insights using **AWS** Quicksight Monitoring Data Quality and integrity end to end testing and reverse engineering and documented existing program and codes.

Environment: Athena, AWS, CI/CD, Cluster, DynamoDB, Elasticsearch, ETL, ER/Studio, GCP, Git, HBase, Jenkins, JS, Kafka, Kubernetes, lake, Lambda, MySQL, PySpark, Python, S3, Snowflake, Spark, SQL, SSIS

IBM || Mumbai, India
GCP Data Engineer || Feb 2021 - Dec 2022

Description: Bell Canada is a Canadian telecommunications company. Used the Cloud Dataflow, Dataproc, or BigQuery for data processing and transformation. Transformed the raw data into structured formats for analytics and reporting.

Responsibilities:

- Strong experience and knowledge of real time data analytics using **Spark** Streaming, Kafka, and **Flume** Configured Spark streaming to get ongoing information from the **Kafka** and store the stream information to **HDFS**.
- Data Ingestion to **Azure Services**, including **Azure** Data **Lake**, **Azure** Storage, **Azure SQL**, and **Azure** DW, as well as data processing in **Azure** Data bricks. Creating Data Studio report to review billing and usage of services to optimize the queries and contribute in cost saving measures.
- Experienced in working **Services** like Data **Lake**, Data **Lake Analytics**, **SQL** Database, Synapse, Data **Bricks**, Data factory, Logic Apps and **SQL** Data warehouse and **GCP** services Like Big Query, Dataproc, Pub sub etc.
- Experienced in **GCP** features which include Google Compute engine, Google Storage, **VPC**, Cloud Load balancing, IAM.
- Created several Databricks **Spark** jobs with **PySpark** to perform several tables to table operations.
- Good knowledge in using Cloud Shell for various tasks and deploying services.

- Developed custom aggregate functions using **Spark SQL** and performed interactive querying.
- Representation of the system in hierarchy form by defining the components, sub components using **Python** and developed set of library functions over the system based on the user needs. Used Cloud Functions to support the data migration from **BigQuery** to the downstream applications. Used **Erwin** for metadata management, improving visibility into data relationships across multiple systems.
- Developed scripts using **PySpark** to push the data from **GCP** to the third-party vendors using their **API framework.**
- Good knowledge in building data pipelines in airflow as a service (composer) using various operators.
- Build a program using **Python** and **Apache beam** to execute it in **cloud Dataflow** and to run Data validation jobs between raw source file and big query tables. Extensive use of cloud shell **SDK** in **GCP** to configure/deploy the services like **Cloud Dataproc** (Managed **Hadoop**), **Google Cloud Storage** and **Cloud Bigquery.**
- Used cloud shell **SDK** in **GCP** to configure the services Data Proc, Storage, **BigQuery**. Performed data analysis and data profiling using complex **SQL** queries on various source systems including **Oracle** 10g/11g and **SQL** Server 2012.
- Integrated **Azure Data Factory** with **Blob** Storage to move data through Databricks for processing and then to **Azure Data Lake** Storage and **Azure SQL** data warehouse. Knowledge on Google Cloud Dataflow and **Apache** Beam.
- Work related to downloading **BigQuery** data into pandas or **Spark** data frames for advanced **ETL** capabilities.
- Used **Sqoop** import/export to ingest raw data into Google Cloud Storage by spinning up Cloud Dataproc cluster.
- Used **Apache Airflow** in **GCP** Composer environment to build data pipelines and used various airflow operators like bash operators, Hadoop operators and **Python** callable and branching operators.

Environment: Airflow, Analytics, Apache, Apache Beam, Azure, Azure Data Lake, BigQuery, Blob, Bricks, Data Factory, ETL, Erwin, Factory, Flume, GCP, HDFS, Kafka, Oracle, PySpark, Python, SDK, Services, Spark, Spark SQL, Spark Streaming, SQL, Sqoop

Anuh Pharma Limited || Mumbai, India
Data Engineer || Jun 2018 - Jan 2021

Description: Anuh Pharma Limited is an India-based bulk drug manufacturing company. Monitored data pipelines and storage solutions for performance, reliability, and cost. Used Cloud Monitoring and Cloud Logging to track system performance and troubleshoot issues. Optimized BigQuery queries and data storage for cost and performance.

Responsibilities:

- Designed and developed Security Framework to provide fine grained access to objects in **AWS S3** using AWS **Lambda**, **DynamoDB**. Involved in data validations and reports using **PowerBI**.
- Involved in loading and transforming large sets of Structured, Semi-Structured and Unstructured data and analyzed them by running **Hive** queries. Processed the image data through the Hadoop distributed system by using Map and Reduce then stored into **HDFS**. Performed the migration of large data sets to Databricks (Spark), create and administer cluster, load data, configure data pipelines, loading data from ADLS Gen2 to Databricks using ADF Pipelines.
- Wrote Spark-Streaming applications to consume the data from **Kafka** topics and write the processed streams to **HBase**.
- Prepared scripts to automate the Ingestion process using **PySpark** and **Scala** as needed through various sources such as **API**, **AWS S3**, **Teradata** and **Redshift**. Provisioned high availability of **AWS EC2** instances, migrated legacy systems to **AWS**, and developed Terraform plugins, modules, and templates for automating **AWS** infrastructure.
- Involved in the entire lifecycle of the projects including Design, Development, and Deployment, Testing and Implementation, and support. Developed **ETL** pipelines between data warehouses using a combination of **Python** and **Snowflake**, SnowSQL, writing **SQL** queries against **Snowflake**.
- Design and configure database, Back-end applications and programs. Managed large datasets using **Pandas** data frames and **SQL**. Worked on scheduling all jobs using **Airflow** scripts using **Python** added different tasks to DAG, LAMBDA.
- Building/Maintaining **Docker** container clusters managed by **Kubernetes Linux**, Bash, **Git**, **Docker**.
- Worked on **CI/CD** tools like **Jenkins**, **Docker** in Devops Team for setting up application process from end-to-end using Deployment for lower environments and Delivery for higher environments by using approvals in between.
- Migrated an existing on-premises application to **AWS**. Used **AWS** services like **EC2** and **S3** for small data sets processing and storage, Experienced in Maintaining the Hadoop cluster on **AWS** EMR

Environment: AWS, Cassandra, CI/CD, Docker, DynamoDB, EC2, EMR, Git, HBase, HDFS, Hive, Java, Jenkins, Kafka, Kubernetes, lake, Linux, Pandas, PySpark, Python, Redshift, S3, Scala, Snowflake, Spark, SQL, Teradata.